Multiple Access and Spread Spectrum

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Wireless Communication IC

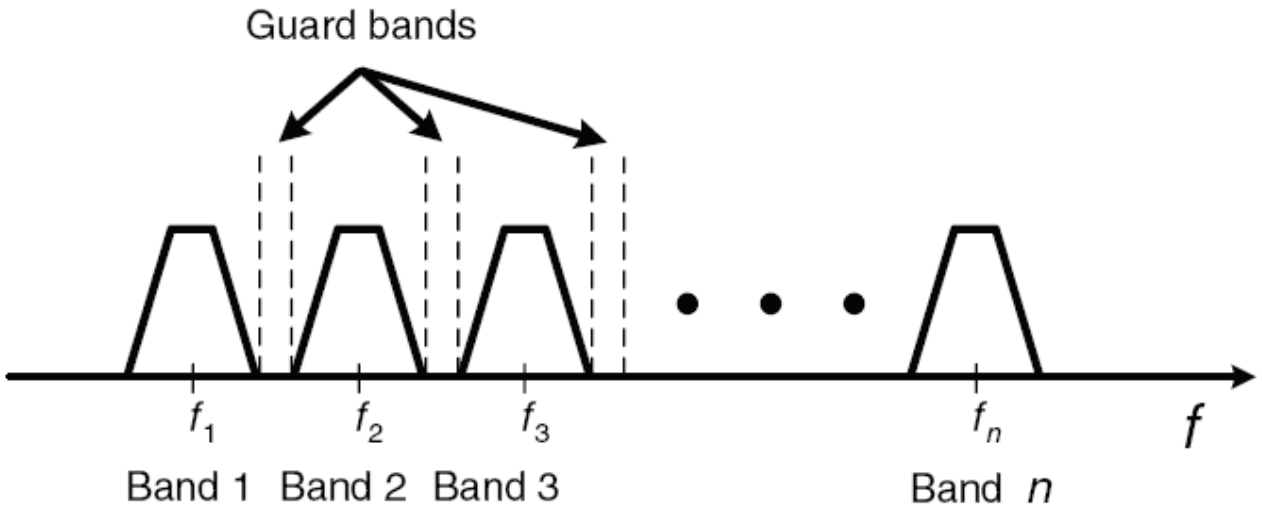
"Baseband Receiver Design for Wireless MIMO-OFDM Communications" 2nd Edition, Wiley 2012.
Chapter 3, Advanced Wireless Technology

Advance Wireless Technology

- Multiple Access
- Spread Spectrum

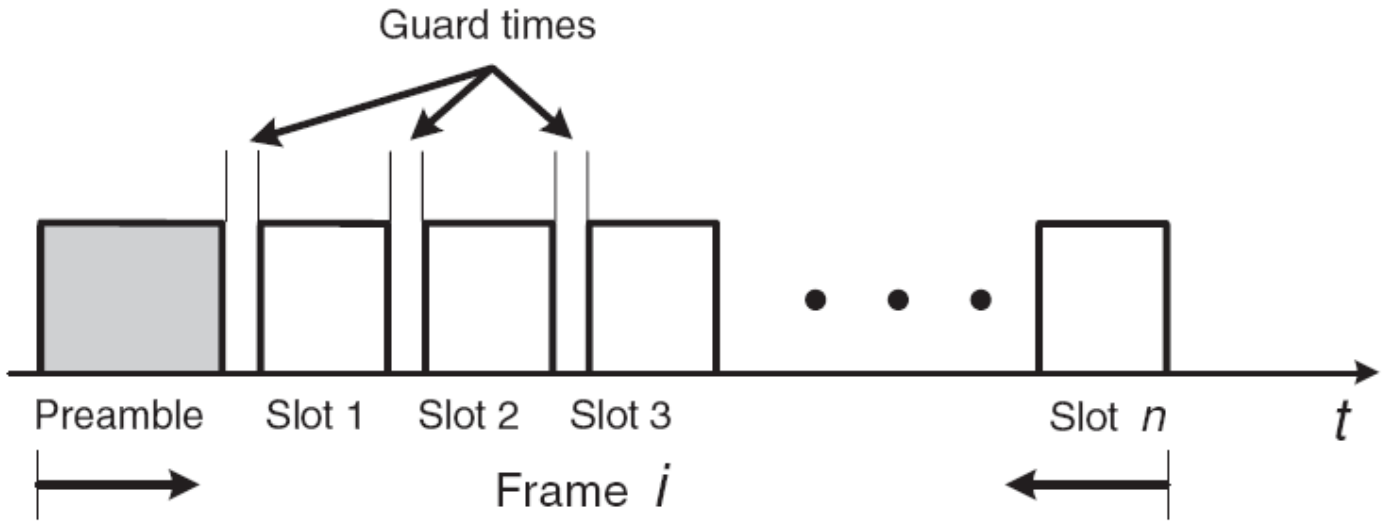
Multiple Access - FDMA

- Frequency division multiple access (FDMA)
  - Allot each frequency band to a user or a group of users
  - Guard bands exist between frequency bands



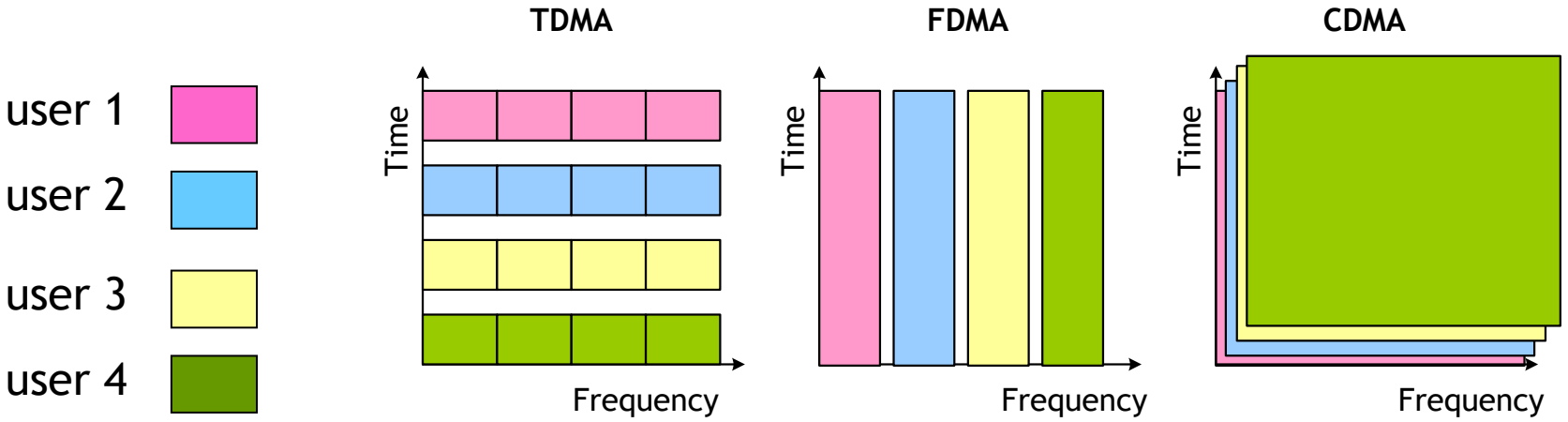
Multiple Access - TDMA

- Time division multiple access (TDMA)
  - Assign a user's signal to a certain time slot
  - Guard time exists between time slots



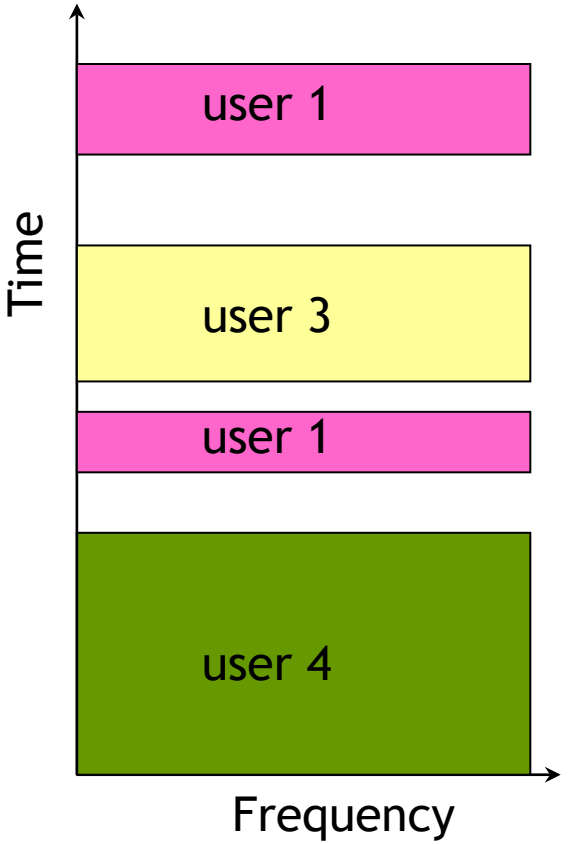
Multiple Access - CDMA

- Code division multiple access (CDMA)
  - All users' signals occupy the same frequency band and the same time slot, but can be distinguished by a user-specific code



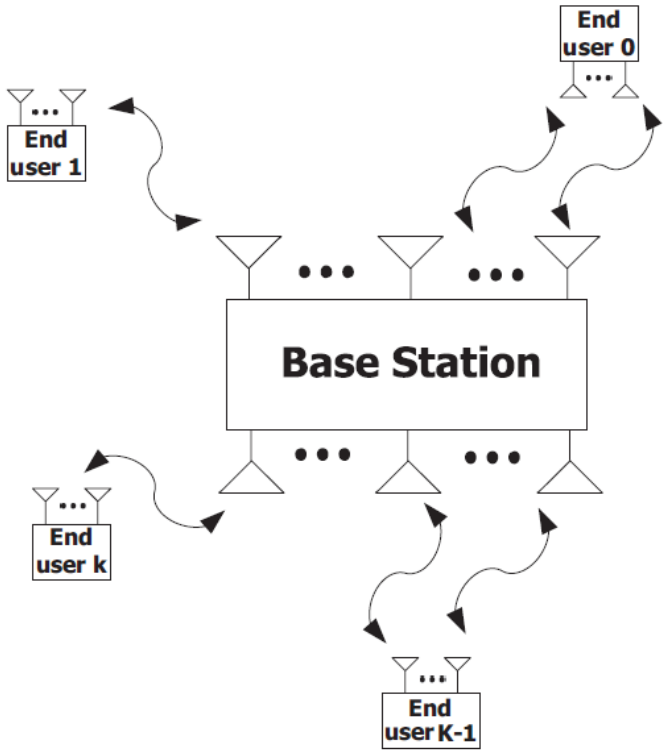
Multiple Access - CSMA

- Carrier sense multiple access (CSMA)
  - Sense before use
  - One user can only send data if the link is clear
  - All users have the same opportunity to access the resources



SDMA

- Spatial division multiple access (SDMA)
  - Similar to precoding in single-user MIMO
  - Spatial interference is hard to avoid
  - Adopted by LTE-A and 802.11ac

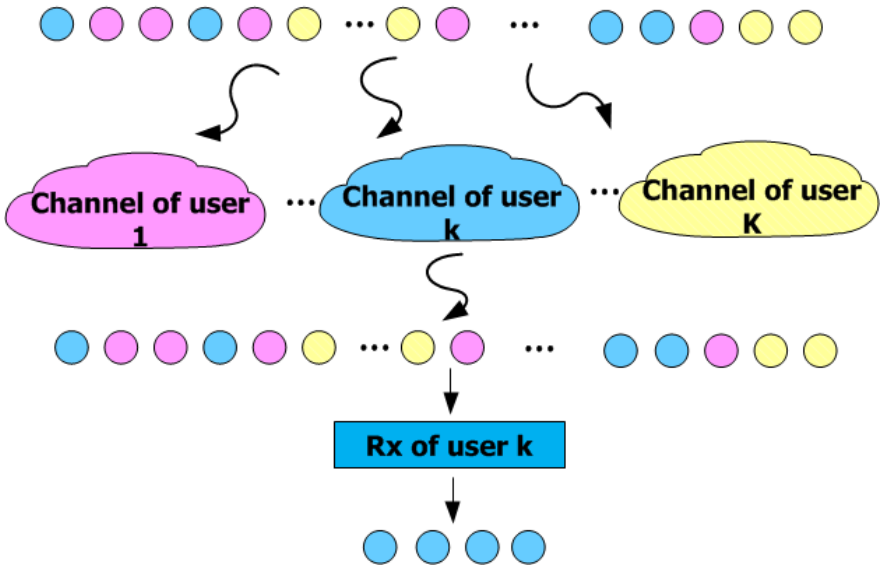


SDMA in Standards

- 2008: 3GPP LTE
- 2013: WiFi 802.11ac
  - 4 multiuser streams in downlink
- 2019: WiFi 802.11ac
  - 8 multiuser streams in downlink and uplink

OFDMA

- Orthogonal frequency division multiple access (FDMA)
  - Different subcarriers are allocated for various users



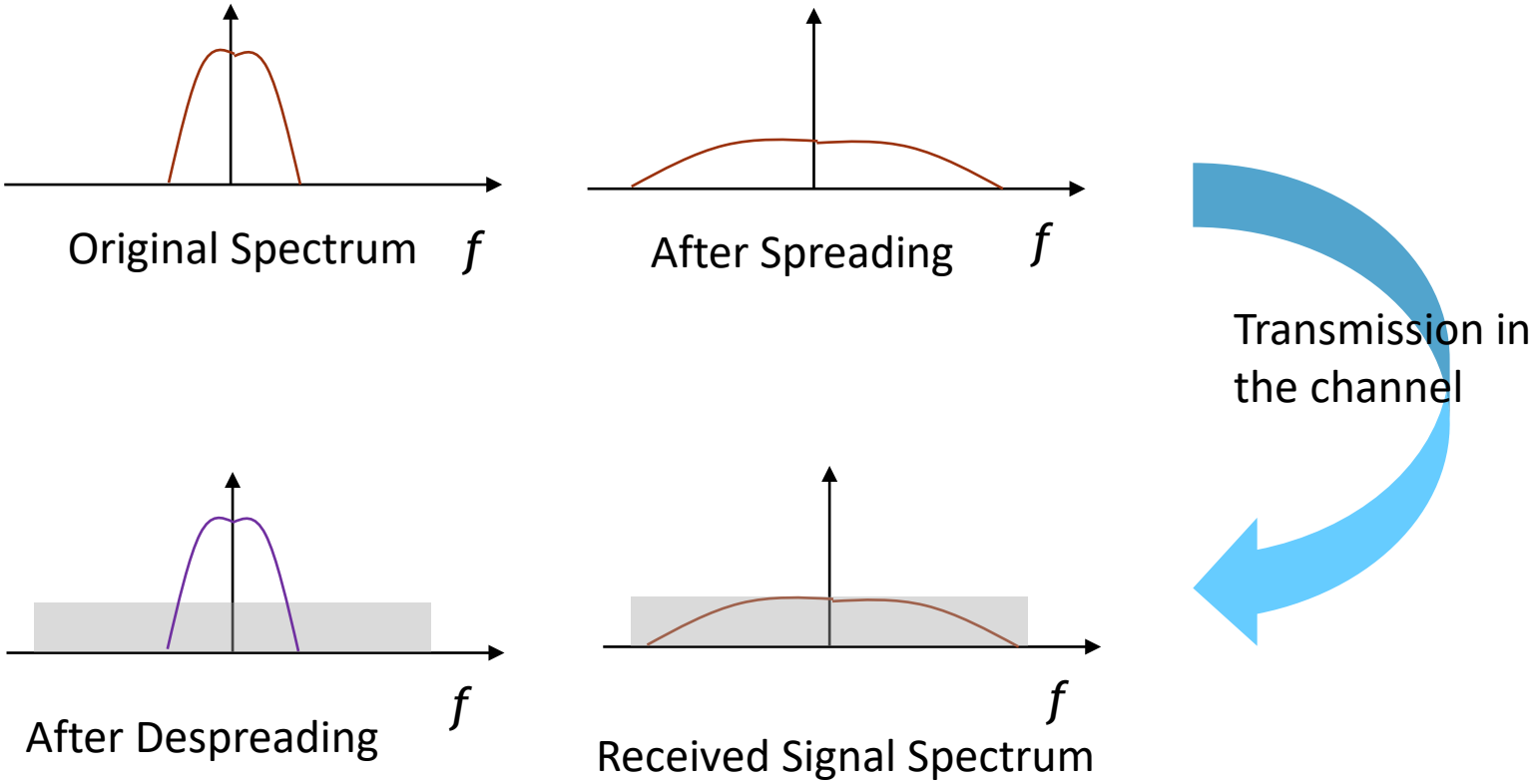
Spread Spectrum and CDMA (1/2)

- Apply a secret code to expand the signal spectrum
- Features of the secret code
  - Noise like
  - Easy to generate
  - Easy to synchronize

Spread Spectrum and CDMA (2/2)

- Advantages of spread spectrum
  - Multipath protection
  - Interference rejection
  - Antijam capability
  - Multiple-access capability
  - Secure communication
  - Improved spectral efficiency

Properties of Spread Spectrum



Spread-Spectrum Techniques

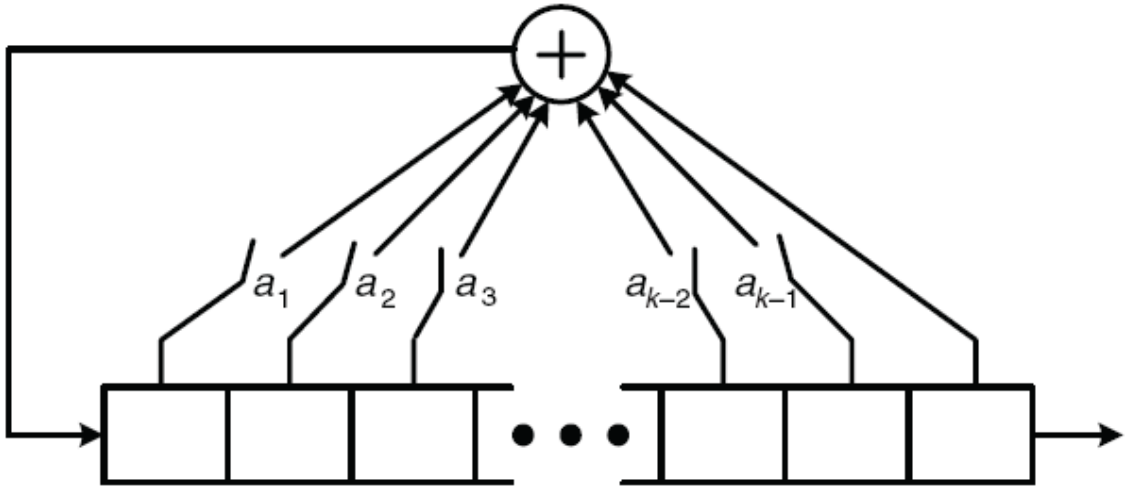
- Pseudo-Random Code (PN Code)
- Direct-Sequence Spread-Spectrum (DSSS)
- Frequency-Hopping Spread-Spectrum (FHSS)

Pseudo-Random (PN) Code (1/2)

- Maximal-length sequences
- Gold codes
- Baker codes

Pseudo-Random (PN) Code (2/2)

- Use Linear Feedback Shift Register (LFSR) to generate a pseudorandom code.
- The coefficients correspond to a primitive root in GF(2^n), where $N = 2^n - 1$.

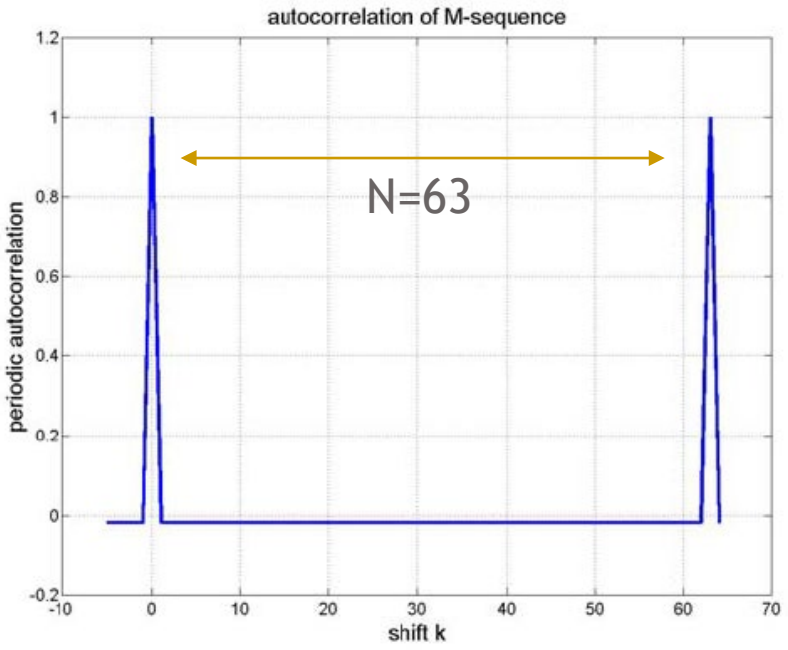


Maximal-Length Sequence

- Contains one more ones than zeroes in a length-*N* sequence.
- Transform (0,1) to (1, -1).
- Periodic autocorrelation function of an *m*-sequence.

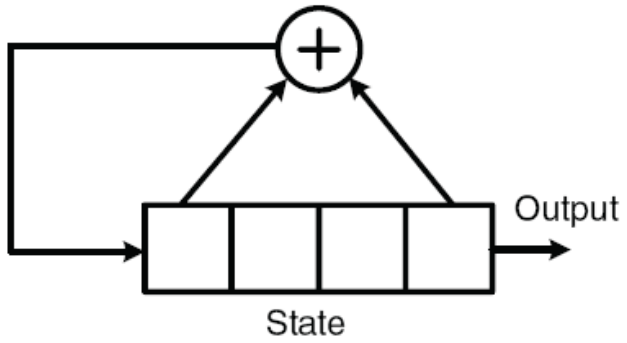
$$\Phi_{bb}(l) = \frac{1}{N}\sum_{n=0}^{N-1} b_n b_{[n+l]_N}$$

$$= \begin{cases} 1 & l = 0 \\ -1/N & l \neq 0 \end{cases}$$



Maximal-Length Sequence Example (1/2)

clock	state	output
1	0001	1
2	1000	0
3	1100	0
4	1110	0
5	1111	1
6	0111	1
7	1011	1
8	0101	1
9	1010	0
10	1101	1
11	0110	0
12	0011	1
13	1001	1
14	0100	0
15	0010	0
16	0001	1

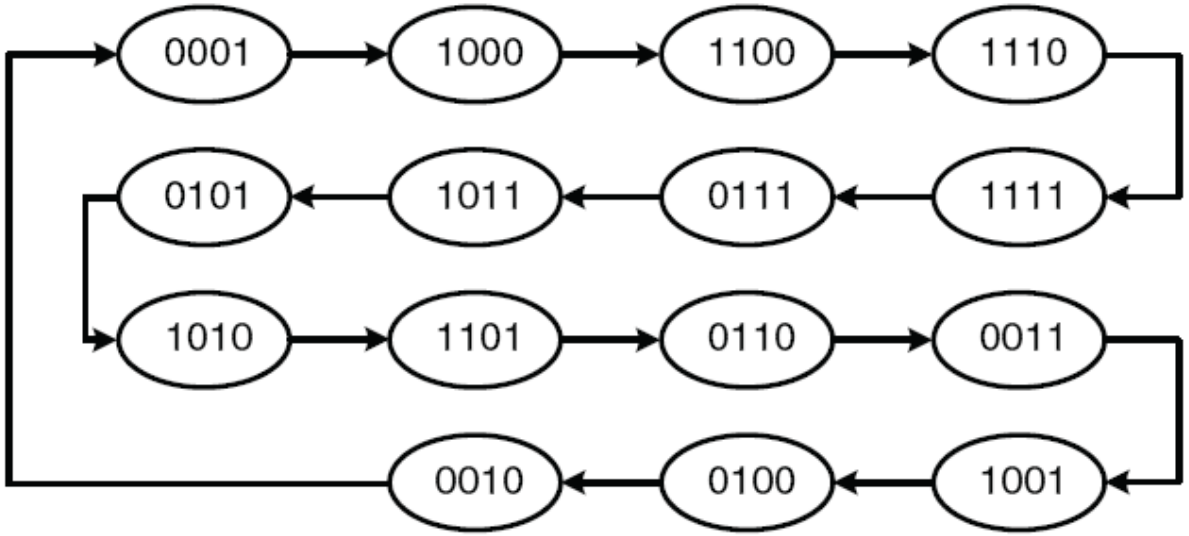


- Primitive polynomial

  x^4+x+1
- $N=2^4-1$

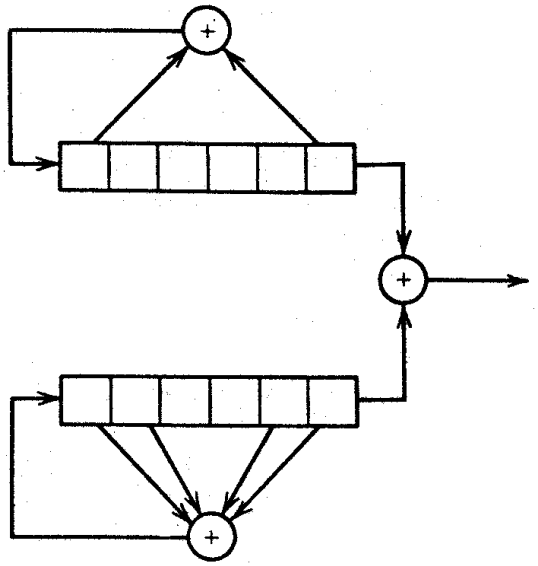
Maximal-Length Sequence Example (2/2)

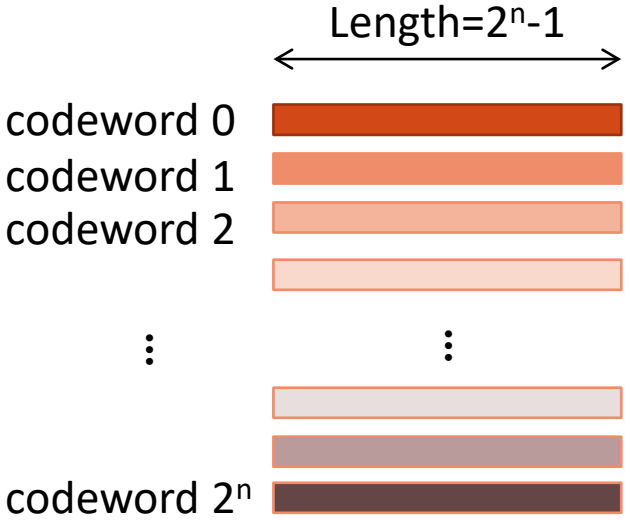
- Out of possible 16 four-bit states, all except the all-zero state are traversed before the state returns to the initial condition.



Gold Codes

- Given two primitive polynomials of degree n forming a preferred pair, $f(x)$ and $g(x)$, one can generate 2^n+1 Gold codes of length 2^n-1 with three possible cross-correlation values





Correlation of Gold Codes

- There are three possible cross-correlation values between any two Gold codes from the same family.
- The auto-correlation values of a Gold code also have the same three values but with different frequencies of occurrence.

$$\Phi_{bb'}(k) = \frac{1}{N}\sum_{n=0}^{N-1} b_n b'_{[n+k]_N}$$

$$\Phi_{bb'}(k) = \begin{cases} -\frac{1}{N}t(n) \\ -\frac{1}{N} \\ \frac{1}{N}(t(n)-2) \end{cases} \qquad t(n) = \begin{cases} 1+2^{0.5(n+1)} & \text{for } n \text{ odd} \\ 1+2^{0.5(n+2)} & \text{for } n \text{ even} \end{cases}$$

Barker Code

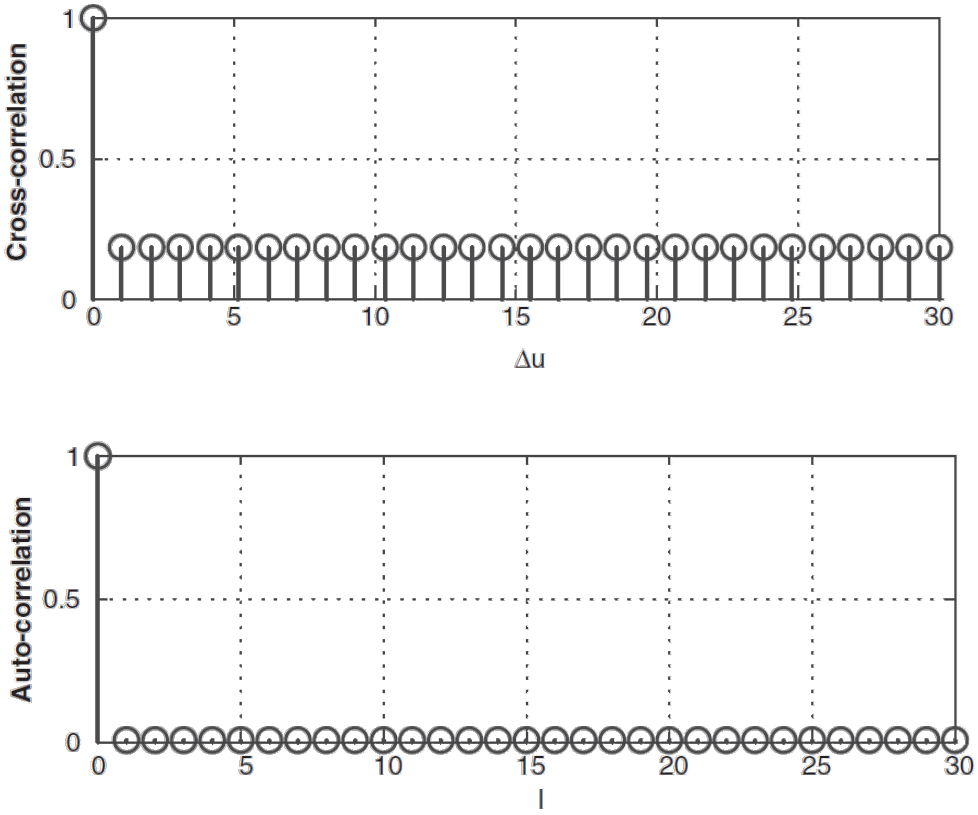
- Popular short PN codes for frame synchronization and direct sequence spread spectrum (DSSS)
- Good auto-correlation property

N	Barker code
7	[−1 −1 −1 1 1 −1 1]
11	[−1 −1 −1 1 1 1 −1 1 1 −1 1]
13	[−1 −1 −1 −1 −1 1 1 −1 −1 1 −1 1 −1]

Zadoff-Chu Sequences

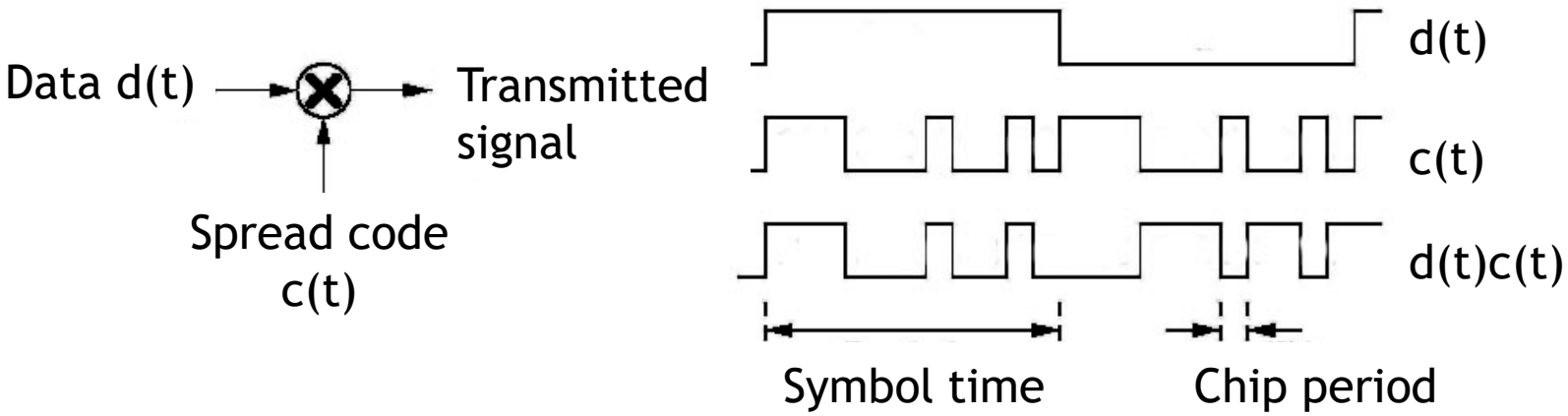
- Complex numbers on the unit circle
- For prime *N*, low cross-correlation among sequences and auto-correlation

$$b_{n,u} = e^{\frac{-j\pi un(n+1)}{N}}$$

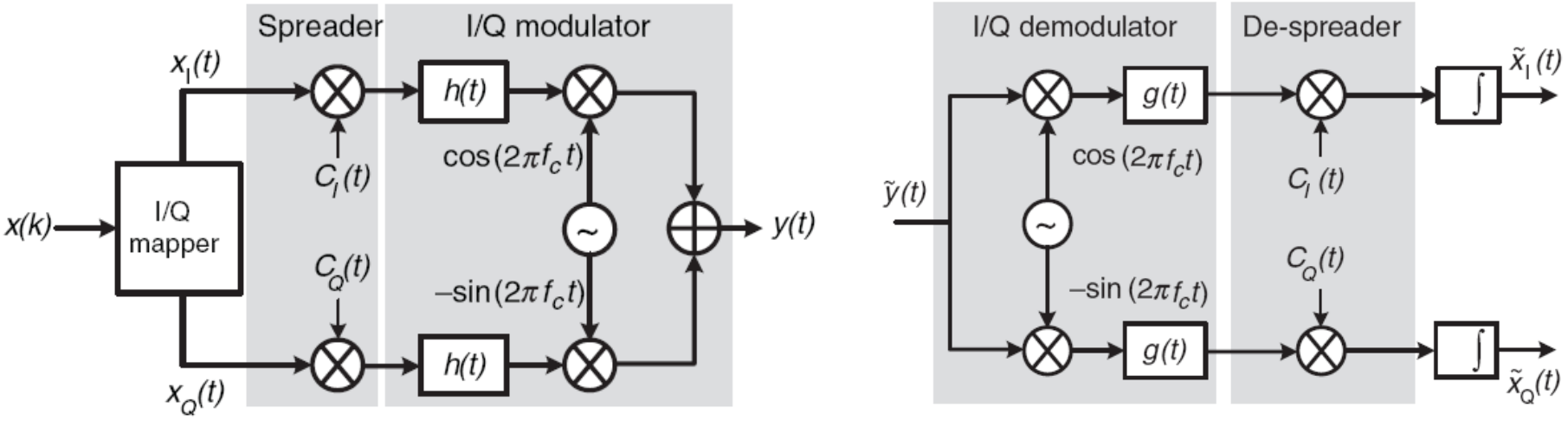


Direct-Sequence Spread Spectrum

- Transmitter
  - Spread the signal spectrum by multiplying a PN code waveform



QPSK DSSS Transceiver



DSSS Signal Waveform

- Only the time-synchronized code sequence can restore the transmitted signals

